

Circuit family $\times$ initialisation $\times$ dataset

A progressive block-size study on DIV2K and TU-Berlin ($m = n = 8, 256 \times 256$)

Generated 2026-05-25 · mean test PSNR over the test split

Summary. We train five quantum-circuit transform families — rich (complex $U(4)$ gates), qft, tebd, entangled_qft, mera — under a block-size curriculum, from two initialisations (identity-operator and random), on two datasets (natural images and sketches). The headline is that **the dataset decides almost everything**: on natural images (DIV2K) only the complex- $U(4)$ rich family clears the classical block-DCT reference and the block-size curve is flat; on block-sparse sketches (TU-Berlin) the curve inverts (small blocks win), the family ranking flips with the compression rate, and classical block-DCT is near-lossless. Across both, **identity init is never worse than random** — often dramatically better.

1 Setup

Each *_progressive sweep trains a circuit family under a **block-size curriculum**: at stage k the inner basis is a $2^k \times 2^k$ circuit replicated across the image by a BlockedBasis wrapper (the bare full circuit at $k = 8$). Stages are independent — no warm-start — and each trains for 1008 steps (generalized preset, 112 epochs) under an MSELoss that keeps the top- k coefficients by magnitude.

Families	rich (complex $U(4)$ 2-qubit gates), qft, tebd, entangled_qft, mera
Initialisations	identity — every gate dropped to its manifold identity ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0), so the circuit starts as the identity operator. random — rich: Haar $U(2)/U(4)$; qft: Haar $U(2)$ on Hadamards + random controlled-phase angles; tebd/entangled_qft/mera: native seeded random gates.
Stage range	rich/qft: $k = 1..8$; tebd/entangled_qft: $k = 2..8$; mera: $k \in \{2, 4, 8\}$ (m must be a power of 2). Tables show $k = 2..8$.
Datasets	DIV2K-8q — 500 natural images, centre-crop + LANCZOS to 256×256 . TU-Berlin — 500 line-drawing sketches, $\approx 96.5\%$ white pixels, same preprocessing.
Metric	mean test PSNR @ keep-ratio ρ (fraction of coefficients retained); $\rho = 0.20$ is the headline rate.

2 DIV2K-8q: natural images

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	32.9	33.7	33.6	33.4	33.4	33.6	33.6
rich	random	32.9	33.4	33.5	32.9	33.2	32.9	33.4
qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
qft	random	32	32.3	31.7	30.9	30.8	31.3	31.3
tebd	identity	32	32.3	32	31.7	31.7	31.7	31.7
tebd	random	32	32.3	31.7	30.9	30.8	30.9	30.9
ent-qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
ent-qft	random	32	32.3	31.7	30.9	31.3	31.3	31.3
mera	identity	32	n/a	32	n/a	n/a	n/a	31.7
mera	random	32	n/a	31.7	n/a	n/a	n/a	30.9

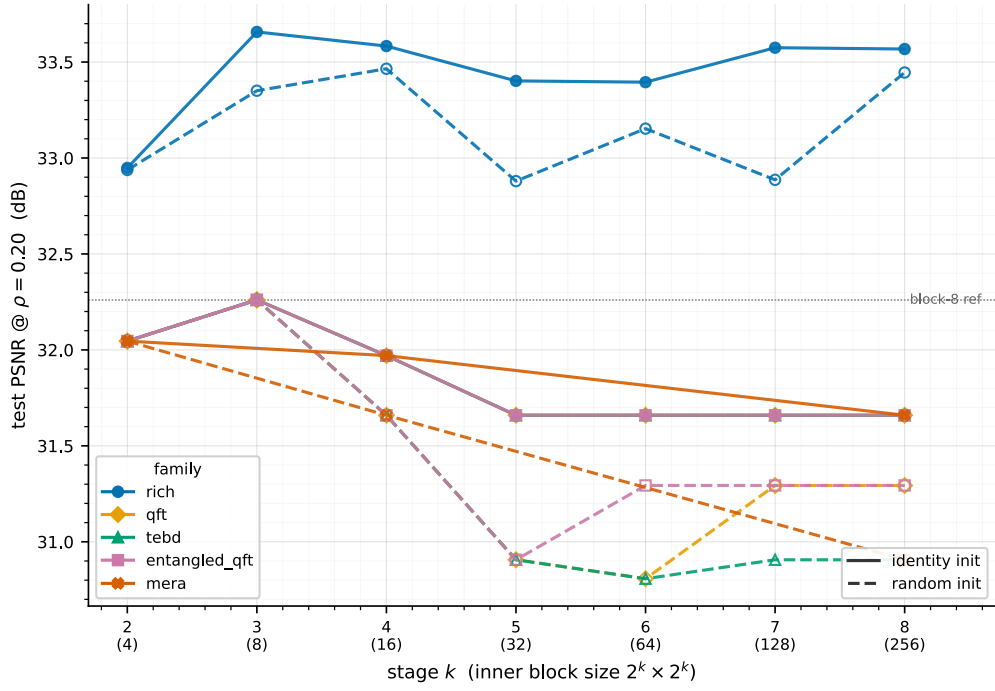


Figure 1: DIV2K, $\rho = 0.20$. Only rich (blue) clears the block-8 reference; the four QFT-derived families coincide under identity init.

Findings.

- **Only rich clears the classical block-8 reference (32.26 dB):** it reaches 33.4–33.7 dB, ≈ 1.5 –2 dB above every other family. The complex $U(4)$ gates are the one thing that buys reconstruction quality here.
- **Under identity init the four QFT-derived families are bit-identical** at every k despite very different gate counts ($k = 8$: qft 72, tebd 32, entangled_qft 80, mera 44). From an identity start they all converge to the same QFT operator – the extra gates train to no-ops.
- **Identity init $\geq$ random** for every family; the curve is essentially flat in k (≈ 33 dB for rich, ≈ 31.7 for the rest).

3 TU-Berlin: sketches (a different regime)

Sketches are **block-sparse** — most 8×8 tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.7 dB mean / 94.5 median at $\rho = 0.20$ (versus ≈ 32 on DIV2K), but only 36.5 dB at $\rho = 0.05$. We therefore report two compression rates.

3.1 Light compression: $\rho = 0.20$ (5 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

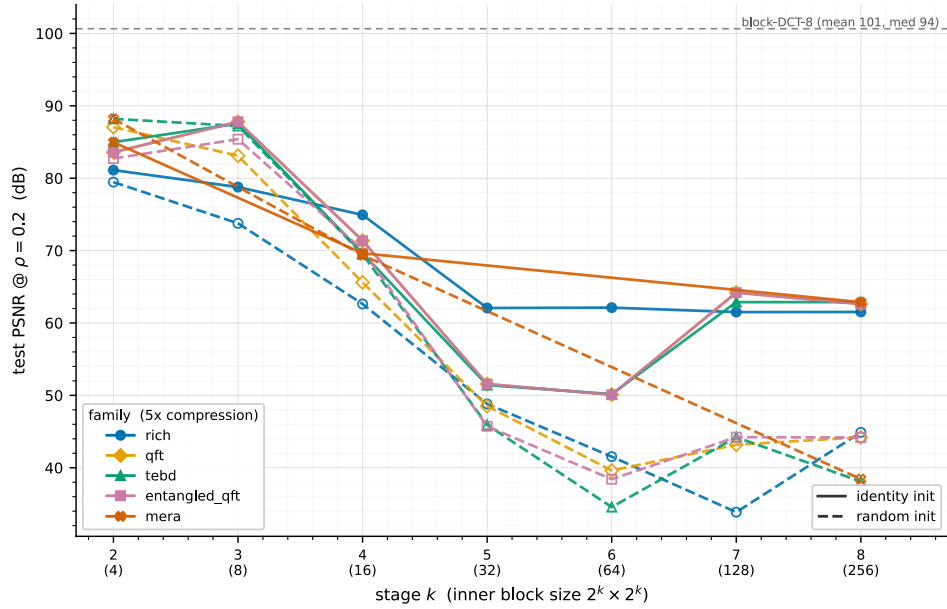


Figure 2: TU-Berlin, $\rho = 0.20$. The curve **falls** with block size; the QFT-derived families peak above rich at $k = 3$; all trail block-DCT-8 (grey).

3.2 Heavy compression: $\rho = 0.05$ (20 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

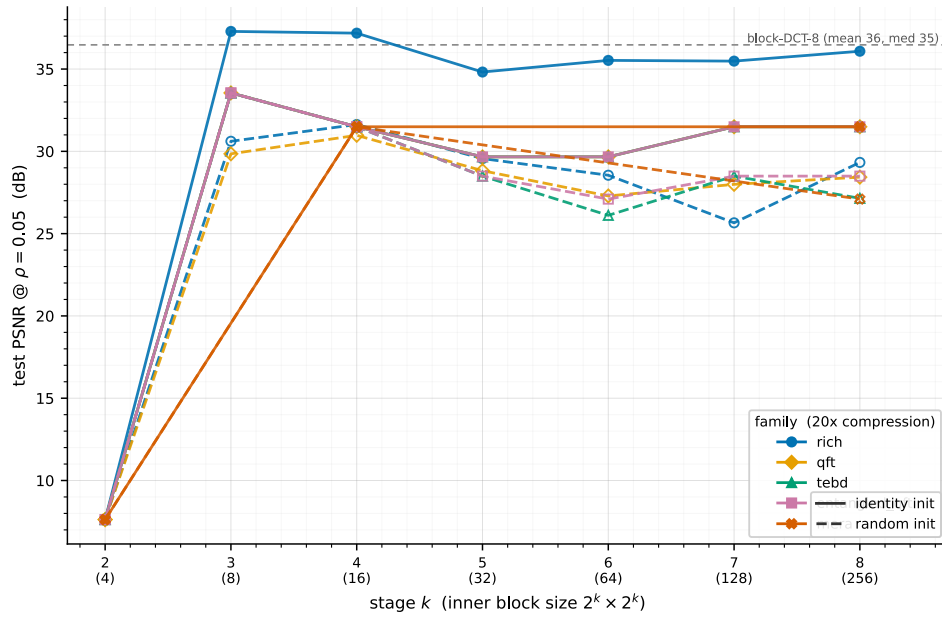


Figure 3: TU-Berlin, $\rho = 0.05$. rich (identity) leads and edges **above** block-DCT-8 at $k = 3, 4$; the per- k curve flattens.

Findings.

- **The block-size curve inverts.** Small blocks win — each blank tile costs almost no coefficients — the opposite of DIV2K’s flat curve.
- **The family ranking flips with the rate.** At $\rho = 0.20$ the QFT-derived families peak ≈ 88 dB at $k = 3$, above rich (≈ 79); at $\rho = 0.05$ rich (identity) leads (≈ 37 dB) and edges above block-DCT-8 (≈ 36).
- **Identity $\triangleright$ random** — the gap reaches ≈ 25 – 30 dB at large k , far wider than on DIV2K. The QFT-family identity equivalence partly survives: $\text{qft} \equiv \text{entangled_qft}$ at all k , and all four QFT-derived families coincide at $\rho = 0.05$.

4 Cross-dataset discussion

	DIV2K (natural)	TU-Berlin (sketch)
block-size curve	flat in k	falls with k (small blocks win)
best learned family	rich (everywhere)	QFT-family @ $\rho=0.2$; rich @ $\rho=0.05$
vs classical block-DCT	rich beats it	trails @ $\rho=0.2$; rich edges it @ $\rho=0.05$
identity vs random	identity $\geq$ random (< 1 dB)	identity $\triangleright$ random (up to 25–30 dB)
absolute PSNR @ $\rho=0.2$	≈ 31 – 34 dB	≈ 60 – 88 dB

The contrast is governed by **energy compaction relative to the data’s sparsity structure**. Natural-image energy is spread across scales, so a global / large circuit and the complex- $U(4)$ richness of rich help, and the curve is flat; sketches are block-sparse, so a small per-block transform already captures almost everything and a fixed block-DCT is near-lossless — learning only pays off once the rate is aggressive enough ($\rho = 0.05$) that even block-DCT degrades.

5 Conclusions

1. **The dataset sets the regime.** Conclusions about “which circuit family is best” do not transfer between natural images and sketches; even the sign of the block-size trend flips.
2. **Complex $U(4)$ (rich) is the only architectural feature that consistently helps** — it wins outright on DIV2K and at heavy compression on sketches.
3. **Identity initialisation dominates random** everywhere, decisively so on sketches; the structured identity start is a strong, cheap prior.

Data: results/family_init_matrix/{div2k_8q,tuberlin_8q}/<family>_<init>/. Reproduce a sweep with experiments/qft_progressive.py --dataset <d> --family <f> --init <i>; regenerate figures with each report’s render_fig.py.